# Comparing Proportions

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This page reviews tests for comparing groups on a categorical outcome: the chi-square test and Fisher's exact test for contingency tables. It uses the chi-square reference distribution defined on the Statistical Inference page. This page is adapted from Vittinghoff et al. (2012), Chapter 3.

## The HERS data

The examples on this page use the HERS data, which the Comparing Means page describes. The `rmb` R package includes the dataset; `haven::as_factor()` converts its Stata value labels to factors:

```
hers <- rmb::hers |> haven::as_factor()
hers |> head()
#> # A tibble: 6 x 37
#>   HT      age raceth nonwhite smoking drinkany exercise physact globrat poorfair
#>   <fct> <dbl> <fct>  <fct>    <fct>   <fct>    <fct>    <fct>   <fct>   <fct>
#> 1 plac~    70 Afric~ yes      no      no       no       much m~ good    no
#> 2 plac~    62 Afric~ yes      no      no       no       much l~ good    no
#> 3 horm~    69 White  no       no      no       no       about ~ good    no
#> 4 plac~    64 White  no       yes     yes      no       much l~ good    no
#> 5 plac~    65 White  no       no      no       no       somewh~ good    no
#> 6 horm~    68 Afric~ yes      no      yes      no       about ~ good    no
#> # i 27 more variables: medcond <dbl>, htnmeds <fct>, statins <fct>,
#> #   diabetes <fct>, dmpills <fct>, insulin <fct>, weight <dbl>, BMI <dbl>,
#> #   waist <dbl>, WHR <dbl>, glucose <dbl>, weight1 <dbl>, BMI1 <dbl>,
#> #   waist1 <dbl>, WHR1 <dbl>, glucose1 <dbl>, tchol <dbl>, LDL <dbl>,
#> #   HDL <dbl>, TG <dbl>, tchol1 <dbl>, LDL1 <dbl>, HDL1 <dbl>, TG1 <dbl>,
#> #   SBP <dbl>, DBP <dbl>, age10 <dbl>
```

## Comparing two groups: categorical outcomes

### Contingency tables

**Example 0.1** (Exercise by treatment group in HERS)**.** Table 1 cross-tabulates regular exercise at baseline by treatment group, as a contingency table with row percentages.

Table 1: Exercise by treatment group in HERS

```
hers |>
  gtsummary::tbl_cross(
    row = exercise,
    col = HT,
    label = list(
      exercise ~ "Exercises regularly",
      HT ~ "Treatment group"
    ),
    percent = "row"
  )
```

| | Treatment group | | |
|---|---|---|---|
| | placebo | hormone therapy | Total |
| Exercises regularly | | | |
| no | 853 (50%) | 842 (50%) | 1,695 (100%) |
| yes | 530 (50%) | 538 (50%) | 1,068 (100%) |
| Total | 1,383 (50%) | 1,380 (50%) | 2,763 (100%) |

### The chi-square test

**Definition 0.1** (Expected count under independence)**.** Let a contingency table have $r$ rows and $c$ columns, with observed count $O_{ij}$ in row $i$ and column $j$, row totals $R_i$, column totals $C_j$, and grand total $n$. The **expected count** in cell $(i, j)$ under independence is

$$E_{ij} \stackrel{\text{def}}{=} \frac{R_i \, C_j}{n}.$$

**Example 0.2** (Expected count in a 2 x 2 table)**.** In a table with $n = 100$, first-row total $R_1 = 30$, and first-column total $C_1 = 40$, the expected count in cell $(1, 1)$ is

$$\begin{aligned} E_{11} &= 30 \cdot 40/100 \\ &= 12. \end{aligned}$$

**Definition 0.2** (Pearson's chi-square test of independence)**.** With observed counts $O_{ij}$ and expected counts $E_{ij}$ in an $r \times c$ contingency table, **Pearson's chi-square test** of the null hypothesis that the row and column variables are independent uses the statistic

$$X^2 \stackrel{\text{def}}{=} \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}.$$

Its p-value is $\Pr(W \geq X^2)$, where $W$ has the $\chi^2_{(r-1)(c-1)}$ distribution (chi-square distribution).

**Theorem 0.1** (Large-sample null distribution of the chi-square statistic)**.** *Let $n$ observations be sampled independently and classified by two categorical variables, and let the two variables be independent. Then as $n \to \infty$, the distribution of $X^2$ (Definition 0.2) converges to the $\chi^2_{(r-1)(c-1)}$ distribution (Hogg et al. 2019, sec. 9.2, p. 440).*

*Remark* 0.1 (Small expected counts). The chi-square approximation is poor when some expected counts are small. A common rule of thumb asks for every $E_{ij}$ to be at least 5.

**Example 0.3** (Chi-square test of exercise by treatment group in HERS)**.** The expected counts and statistic of Definition 0.2 for Table 1:

```r
observed <- table(hers$exercise, hers$HT)
expected <- outer(rowSums(observed), colSums(observed)) / sum(observed)
x_squared <- sum((observed - expected)^2 / expected)
expected
#>     placebo hormone therapy
#> no   848.42          846.58
#> yes  534.58          533.42
c(
  x_squared = x_squared,
  p_value = pchisq(x_squared, df = 1, lower.tail = FALSE)
)
#> x_squared   p_value
#>  0.128054  0.720458
```

Every expected count is large, so the $\chi^2_1$ approximation of Theorem 0.1 is reasonable. `chisq.test()` with `correct = FALSE` reports the same values:

```r
chisq.test(hers$exercise, hers$HT, correct = FALSE)
#>
#>  Pearson's Chi-squared test
#>
#> data:  hers$exercise and hers$HT
#> X-squared = 0.1281, df = 1, p-value = 0.72
```

The p-value is large: the data give no evidence that exercise depends on treatment group, as randomization would lead us to expect.

*Remark* 0.2 (Yates' continuity correction). For a $2 \times 2$ table, `chisq.test()` applies Yates' continuity correction by default, which subtracts 0.5 from each $|O_{ij} - E_{ij}|$ before squaring, and so gives a smaller statistic than Definition 0.2. `correct = FALSE` turns the correction off.

### Fisher's exact test

**Definition 0.3** (Fisher's exact test)**.** Take a $2 \times 2$ contingency table with cells $a$, $b$, $c$, and $d$ as in the contingency table definition, and hold its row and column totals fixed. Under independence of the row and column variables, the probability that the top-left cell equals $x$ is the hypergeometric probability

$$p(x) \stackrel{\text{def}}{=} \frac{\binom{a+b}{x}\binom{c+d}{a+c-x}}{\binom{n}{a+c}}.$$

**Fisher's exact test** of independence has p-value

$$\sum_{x\,:\,p(x)\le p(a)} p(x),$$

the total probability of the tables with the same totals that are no more probable than the observed table.

*Remark* 0.3 (Why Fisher's exact test is exact). The p-value is exact: it comes from the null distribution itself, not from a large-sample approximation, so the test is valid even when expected counts are small, and it is often used for $2 \times 2$ tables in which some expected count is below 5 (Theorem 0.1). Other two-sided versions exist; this one is the version that R's `fisher.test()` computes.

**Example 0.4** (Fisher's exact test of exercise by treatment group in HERS)**.** The p-value of Definition 0.3 for Table 1, computed from the hypergeometric probabilities with `dhyper()`:

```
a <- observed[1, 1]
row1 <- sum(observed[1, ])
row2 <- sum(observed[2, ])
col1 <- sum(observed[, 1])
x <- max(0, col1 - row2):min(row1, col1)
p_x <- dhyper(x, m = row1, n = row2, k = col1)
p_a <- dhyper(a, m = row1, n = row2, k = col1)
sum(p_x[p_x <= p_a * (1 + 1e-7)])
#> [1] 0.72528
```

The tolerance `1e-7` keeps tables whose probability equals $p(a)$ up to rounding error, as `fisher.test()` does. `fisher.test()` reports the same p-value:

```
fisher.test(hers$exercise, hers$HT)
#>
#>  Fisher's Exact Test for Count Data
#>
#> data:  hers$exercise and hers$HT
#> p-value = 0.725
#> alternative hypothesis: true odds ratio is not equal to 1
#> 95 percent confidence interval:
#>  0.879664 1.202192
#> sample estimates:
#> odds ratio
#>    1.02836
```

With counts this large, the exact p-value is close to the chi-square p-value of Example 0.3.

### Measures of association for $2 \times 2$ tables

Tests of independence say whether two binary variables are associated, but not how strongly. Risk differences, risk ratios, and odds ratios measure the strength of the association; see Odds Ratios and Relative Risks.

## References


Hogg, Robert V., Elliot A. Tanis, and Dale L. Zimmerman. 2019. *Probability and Statistical Inference*. Tenth edition. Pearson.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. https://doi.org/10.1007/978-1-4614-1353-0.